Biliary Atresia, Part Thirteen: The Adult With Biliary Atresia. Transition to Adult Care, Pregnancy, and Hepatocellular Carcinoma and Cholangiocarcinoma in the Native Liver.

Imagine a nineteen-year-old woman who had a Kasai operation for biliary atresia at forty-eight days of age. Her jaundice cleared in infancy, and she has grown up with her own liver. Today her bilirubin is one point six milligrams per decilitre, just above the upper limit of normal. Her platelet count is ninety-two thousand per microlitre, and her spleen measures seventeen centimetres. She had two episodes of cholangitis when she was fifteen. She is about to move from the children's hospital to an adult liver clinic, and at her last paediatric visit she says that she and her partner would like to have a baby. The questions for the clinician are these. What does her future liver course look like, and how likely is she to need a transplant as an adult? How should the hand-over to adult care be done so that nothing is lost? Is pregnancy safe for her, and how should it be planned? And does her native liver carry a risk of cancer that someone needs to watch for?

The main idea of this chapter is that biliary atresia does not end when the jaundice clears. A successful Kasai turns a fatal disease of infancy into a chronic liver disease of adulthood. Almost every adult who still has a native liver has biliary cirrhosis, and the questions change as the patient grows up. They are no longer about surviving infancy but about the timing of a late transplant, the safety of pregnancy, surveillance for cancer, and a careful hand-over to adult physicians who may never have looked after a patient with a Kasai.

A few terms first. The Kasai operation, also called a hepatic portoenterostomy, removes the scarred remnant of the bile ducts and joins a loop of small bowel directly to the liver at the porta hepatis, the point where the ducts would normally leave the liver. The loop is called a Roux loop, or a Roux-en-Y loop, because the bowel is arranged in the shape of a letter Y so that food does not pass up into the limb attached to the liver. Native-liver survival means being alive without having had a transplant. Portal hypertension is raised pressure in the portal vein, the vein that carries blood from the gut to the liver; it causes an enlarged spleen, a low platelet count, and varices, which are swollen veins in the oesophagus and stomach that can bleed. Hepatocellular carcinoma, shortened here to HCC, is cancer of the liver cells. Cholangiocarcinoma is cancer of the bile duct lining. Alpha-fetoprotein, or AFP, is a blood marker for hepatocellular carcinoma, and carbohydrate antigen nineteen-nine, usually written CA nineteen-nine, is a blood marker for biliary cancer.

The evidence in this chapter comes from long-term cohorts and small series, because there are no trials in adults with biliary atresia. The French national cohort was reported by Fanna and Chardot in 2019, and the Japanese Biliary Atresia Registry was reported by Okubo and colleagues in 2025. The classic description of adults twenty years after a Kasai is the Paris series by Lykavieris and colleagues, published in Hepatology in 2005. Predictors of an adult transplant come from the King's College Hospital study by Jain and colleagues in the Journal of Hepatology in 2019, and adult transplant outcomes come from the King's surgical series by Cortes-Cerisuelo in 2021 and a United States registry analysis by Moazzam in 2021. Pregnancy data come from Japanese series by

Shimaoka in 2001, Sasaki in 2007 and Kuroda in 2007. Cancer data come from the King's experience reported by Hadzic in 2011 and a 2025 systematic review by Madadi-Sanjani and colleagues. Guidance comes from the 2018 joint position paper on transition from the European Society for Paediatric Gastroenterology, Hepatology and Nutrition and the European Association for the Study of the Liver, the 2021 reproductive health guidance from the American Association for the Study of Liver Diseases, the 2023 European Association for the Study of the Liver guideline on liver disease in pregnancy, and the adult guidelines on hepatocellular carcinoma surveillance. For brevity, these societies will be called the American association, the European association, and the European paediatric society.

Part one. Why the native liver keeps ageing

To understand the adult with biliary atresia, it helps to think of five processes that keep running after the Kasai, even when it has worked.

The first is fibrosis. Biliary atresia is a progressive inflammatory disease of the whole biliary tree, not just a blocked duct. The Kasai restores drainage from the tiny ductules at the porta, but the injury to the intrahepatic ducts often continues slowly. Over years this produces cirrhosis. In the Paris series, nearly every patient who was alive with a native liver twenty years after surgery had signs of cirrhosis.

The second is the Roux loop itself. Because the bile ducts drain into a loop of bowel, bacteria can travel up into the liver. This is ascending cholangitis. Over time, dilated pockets of bile called bile lakes, and stones inside the liver, can form and become sources of repeated infection. The practical consequence for adult doctors is very important: there is no papilla for an endoscope to enter, so the usual adult tool for draining the bile ducts, endoscopic retrograde cholangiopancreatography, usually shortened to ERCP, cannot reach these ducts.

The third is portal hypertension. Many children with biliary atresia have a small portal vein, and cirrhosis raises the resistance to portal flow. The result is varices, an enlarged spleen, a low platelet count and, in adults, large spontaneous shunts between the portal and systemic veins. The splenic artery often enlarges, and some patients develop aneurysms of the splenic artery.

The fourth is lung vascular disease. Over years, abnormal vessels can open in the lungs, causing hepatopulmonary syndrome, in which blood bypasses the air spaces and oxygen levels fall, or the opposite problem, portopulmonary hypertension, in which the pulmonary arteries narrow and the right heart is strained. Both can appear late, in patients who otherwise seem well.

The fifth is neoplasia. Constant regeneration in a cholestatic, cirrhotic liver gives a setting for hepatocellular carcinoma, and long-standing injury to the bile duct lining gives a setting for cholangiocarcinoma.

A helpful way to hold this together is to imagine two clocks. The synthetic clock, which governs how well the liver makes albumin and clotting factors, runs slowly in biliary atresia. Albumin and the international normalised ratio, or INR, can stay normal for decades. The portal hypertension clock and the biliary clock run faster, and they are the ones that bring adults to transplantation. Scores built for adult cirrhosis read mainly the slow clock, and so they tend to miss the danger in

these patients.

Part two. How many reach adulthood with their own liver

The French national cohort included one thousand three hundred and forty children who had a Kasai between 1986 and 2015. Survival with the native liver was thirty-five percent at ten years, twenty-six percent at twenty years, and twenty-two percent at thirty years. Timing of the Kasai mattered: twenty-five-year native-liver survival was thirty-eight percent when the operation was done in the first month of life, and nineteen percent when it was done after the third month. Overall, about eighty-seven percent of French children with biliary atresia now survive, most of them after a transplant.

The Japanese registry, with nearly four thousand patients, reports higher figures: native-liver survival of about fifty point five percent at ten years, forty-four percent at twenty years, and forty-one percent at thirty years. When these numbers were drawn as bars, the French bars fell from about a third to about a fifth, while the Japanese bars stayed between about a half and two fifths. The difference partly reflects era, earlier diagnosis and screening with stool colour cards in Japan, and differences in transplant access, so the registries are not directly comparable. But the message is the same in both. Between one fifth and two fifths of every birth cohort will reach adulthood on their own liver, and every adult hepatology unit will meet these patients in growing numbers.

Part three. What adulthood looks like

The Paris series by Lykavieris followed two hundred and seventy-one children operated on between 1968 and 1983. Twenty years later, sixty-three of them, or twenty-three percent, were alive with their native liver. This is the best portrait we have of what a good Kasai result looks like in adulthood, and it is sobering. Sixty-one of the sixty-three had signs of cirrhosis. Only twenty-one had a normal bilirubin, and only twelve had normal bilirubin, normal aminotransferases and a normal gamma-glutamyl transferase, the enzyme that rises in bile duct injury. Forty-four had portal hypertension, nineteen had late bacterial cholangitis, and six had gallstones. After the age of twenty, fourteen had a transplant or were on the waiting list, and two died of liver failure. The authors estimated that fewer than eighteen percent of infants treated with a Kasai would avoid a transplant over their lifetime.

There is a hopeful side to this portrait too. Seven women in the series had nine children, and three men fathered six. In a Japanese series reported by Kuroda, forty-five of forty-seven adults who survived beyond twenty years, or almost ninety-six percent, were employed, in higher education, or married. These patients live full lives. But a normal bilirubin is not a normal liver, and they need lifelong specialist follow-up.

Part four. What at sixteen predicts a transplant later

The most useful study for the transition clinic comes from King's College Hospital in London. Jain and colleagues studied three hundred and ninety-seven patients who had a Kasai between 1980 and 1996. One hundred and eleven, or twenty-eight percent, still had their own liver at sixteen. Of those followed beyond sixteen, sixty-seven kept their liver and twenty-two needed a transplant as adults. The researchers looked at what had been measured around the sixteenth birthday.

A higher bilirubin at sixteen predicted a later transplant. A bilirubin of twenty-one micromoles per litre or more, which is about one point two milligrams per decilitre and only just above the upper limit of normal, predicted an adult transplant with an area under the curve of zero point eight five, a sensitivity of eighty-five percent and a specificity of seventy-four percent. Cholangitis during adolescence carried about a five-fold higher risk of later transplant. Portal hypertension before sixteen carried about a seven-fold risk, and varices before sixteen carried an eight point six-fold risk. When these were shown as bars, the bars grew from cholangitis, to portal hypertension, to varices, with varices the strongest. Interestingly, a lower creatinine at sixteen also predicted a later transplant. Creatinine comes from muscle, so a low value probably reflects low muscle mass, a marker of sarcopenia and chronic illness.

The practical lesson is that a bilirubin barely above normal at sixteen is not reassuring. It is the single best predictor of an adult transplant. The transfer letter should state the bilirubin at sixteen, whether there are varices, and how many episodes of cholangitis there have been in adolescence. The authors also noted that adult scores such as the Model for End-Stage Liver Disease, or MELD, were not designed for this population and miss the biliary complications that drive its course.

Part five. Transition to adult care

Transition is the planned, gradual process of preparing a young person to manage their own health and to move from a paediatric to an adult service. Transfer is the single event of moving. The joint position paper from the European paediatric society and the European association, published by Vajro and colleagues in 2018, and the later review by Kelly, Samyn and Schwarz, describe the essential elements.

Preparation should start in early adolescence, at around twelve years of age. There is no validated readiness tool and no agreed age of transfer, so readiness is judged clinically. The young person should be able to name the diagnosis, describe the operation, list their medicines, and recognise warning signs. For biliary atresia this matters especially, because fever with jaundice may be cholangitis and black stools or vomiting blood may be variceal bleeding, and both need urgent care.

The transfer itself should happen when the patient is stable. It should never happen during an episode of cholangitis, a variceal bleed, or, in transplant recipients, an episode of rejection or graft dysfunction. A joint clinic, in which the paediatric and adult hepatologists see the patient together, is the ideal way to hand over the anatomy and the history. The written transfer summary should include the date of the Kasai, the anatomy of the Roux loop, the history of varices and their treatment, the number of cholangitis episodes, and the most recent imaging. Adult teams rarely see biliary atresia, and they cannot be expected to reconstruct this history from scratch.

Adherence deserves particular attention. Late graft loss after transplantation clusters in adolescents and young adults, and missed appointments and missed medicines are a major reason. The transition period is also the time to screen for anxiety and depression, to discuss alcohol, and to support education and employment. Finally, contraception and pregnancy counselling should be part of transition, before the transfer, because pregnancies in young adults are often unplanned. The best way to think of transfer is as an elective procedure: planned, documented, and never done in an emergency.

Part six. What the adult hepatologist must know

Several features of biliary atresia are not anticipated by adult protocols, and they are worth stating plainly.

The first and most important is the Roux loop. An adult with fever and jaundice after a Kasai looks, at first sight, like an adult with cholangitis from a stone, and the reflex in adult medicine is to ask for an ERCP. In biliary atresia this is usually impossible, because the ducts drain into a loop of bowel far from the papilla. The right approach is antibiotics, blood cultures, imaging with ultrasound and magnetic resonance cholangiopancreatography, called MRCP, and if needed drainage of a bile lake or an obstructed segment by an interventional radiologist through the skin. Recurrent cholangitis should be treated seriously, with a prolonged course of antibiotics and consideration of prophylaxis, because every episode adds to the damage.

The second is that normal clotting and albumin do not mean low risk. Because the MELD score is built from bilirubin, INR and creatinine, an adult with biliary atresia can have a MELD of nine or ten while suffering from recurrent cholangitis and large varices. The decision to refer for transplantation should be driven by complications, not by the score.

The third is vascular. Splenic artery aneurysms are common in adults with long-standing portal hypertension, and they can rupture, especially in pregnancy and during abdominal surgery. Cross-sectional imaging before pregnancy or surgery is wise.

The fourth is the lungs. Pulse oximetry should be checked at every annual review, and an echocardiogram should be done if there is breathlessness or low oxygen saturation, because hepatopulmonary syndrome and portopulmonary hypertension can appear late. Finally, bone health and fat-soluble vitamins should not be forgotten. Vitamin D deficiency is common, and fractures occur in native-liver survivors.

Part seven. Adults wait longer and die more often

When an adult with biliary atresia needs a transplant, the system is often less kind than it was in childhood. Moazzam and colleagues analysed the Scientific Registry of Transplant Recipients in the United States from 2002 to 2018. They compared one hundred and fourteen adolescents with two hundred and seventeen adults listed for biliary atresia. As summarised in the 2023 review by Kelly and colleagues, adults had about eleven times the risk of dying on the waiting list on multivariable analysis, and five-year patient survival after transplant was about eighty-four percent in adults compared with ninety-eight percent in adolescents.

At King's College Hospital, the median wait for adults was two hundred and ninety-two days, and the MELD score rose from a median of twelve at listing to fifteen at transplant. In Japan, where deceased donation is rare, a single-centre report, available only as a preprint, described six adults listed for a deceased donor liver, all of whom died while waiting, after a median of seventeen months.

Why do adults do worse? There are three reasons. They lose the priority given to children in allocation. Their MELD score is low, because the disease is driven by bile ducts and portal pressure

rather than by synthetic failure. And their parents, who were the natural living donors in infancy, are now older or no longer suitable. On top of this, referral is often delayed across the gap between paediatric and adult services. The lesson is to refer early, to list when complications appear rather than when the MELD rises, and to request exception points when the score understates the risk.

Part eight. The operation in an adult

The King's surgical series by Cortes-Cerisuelo describes thirty-four patients who received a transplant for biliary atresia at sixteen years or older, with a median age of twenty-four, between 1989 and 2020. The indications were synthetic failure in thirty-two percent, recurrent cholangitis in twenty-nine percent, progressive jaundice in eighteen percent, and smaller numbers for variceal bleeding, encephalopathy, hepatopulmonary syndrome and suspected cancer.

The operation is demanding. The abdomen is full of adhesions from the Kasai and often from later operations. Spontaneous shunts between the portal and systemic circulations were found in seventy-three percent, the splenic artery was enlarged in fifty-eight percent, and twenty-one percent had multiple splenic artery aneurysms, some over five centimetres. The portal vein was thin on imaging in almost half. The median operation lasted four hundred and sixty minutes, there was injury to the bowel in about nine percent, and biliary complications occurred in about twenty-one percent.

And yet the results were excellent. Graft survival was ninety-seven point one percent at one year and ninety-one point two percent at ten years. Patient survival was one hundred percent at one year and ninety-four percent at ten years. When drawn as bars, all four bars reached above ninety percent. The message is that in an experienced centre the adult operation rivals any other adult indication. The problem is getting to the operation in time, not surviving it.

Part nine. Pregnancy: what the series show

Pregnancy in women with biliary atresia is possible and usually successful, but it is rarely uneventful. The largest early report is a survey of the Japanese Biliary Atresia Society by Shimaoka and colleagues in 2001. Fourteen institutions reported sixteen women after a Kasai, with twenty-three deliveries. Liver function worsened after delivery in six of the sixteen, or thirty-seven and a half percent. Four, or a quarter, developed ascending cholangitis. Only three, fewer than one in five, had no problems at all. One pregnancy was lost after massive bleeding from oesophageal varices, and one woman needed a transplant because of liver dysfunction after the birth. None of the babies had congenital abnormalities. When these were drawn as bars, the tallest bar was for worsening liver function and the shortest was for a completely uneventful course.

A second series from Sendai, reported by Sasaki and Nio in 2007, looked at fifty-five long-term survivors without transplantation. Nine had been pregnant, with fourteen pregnancies and eleven deliveries. None had visible jaundice before pregnancy, but six had complications such as cholangitis or portal hypertension, two became jaundiced after pregnancy, and there was one intrauterine fetal death. The authors concluded that cholangitis and portal hypertension are risk factors but not contraindications.

Kuroda and colleagues looked at the same question from another angle. Of thirty-three adults with good liver function at twenty, six, or about eighteen percent, developed liver dysfunction after pregnancy or after strenuous work. These six had a lower serum cholinesterase at age twenty than the stable patients. Cholinesterase is an enzyme made by the liver, and a low or falling level is a sensitive marker of declining synthetic reserve. The lesson from these series is that pregnancy is a stress test of the liver. Cholangitis and portal hypertension before conception predict trouble, and deterioration can appear months after delivery.

Part ten. Why pregnancy loads the portal system

The physiology explains why. During pregnancy, plasma volume and cardiac output rise by up to half, and portal blood flow rises with them. The volume peaks in the late second trimester. As the uterus grows, it compresses the inferior vena cava, pushing more blood into collateral veins, including varices. In labour, repeated pushing in the second stage raises intra-abdominal and variceal pressure. Oestrogen reduces bile flow, and stasis in a Roux loop invites cholangitis. And there is a rare but catastrophic risk: splenic artery aneurysms can rupture in pregnancy, with very high mortality for mother and baby.

The 2023 European association guideline summarises the numbers in women with cirrhosis. Variceal bleeding complicates up to a third of pregnancies, and up to half when portal hypertension is already known. Maternal mortality has fallen from as much as fourteen percent in older series in the 1980s to under two percent in recent series. Live birth rates range from about sixty to one hundred percent.

Part eleven. Scoring the risk before conception

The best time to assess a pregnancy is before it starts. The European association guideline recommends pre-pregnancy counselling and calculation of risk scores. In a study by Westbrook and colleagues from King's College Hospital in 2011, a MELD score of six or less before conception was associated with excellent outcomes, while a MELD above ten predicted decompensation during pregnancy. A platelet count above one hundred and fifty thousand together with a liver stiffness below twenty kilopascals makes high-risk varices unlikely, while a platelet count below about one hundred and ten thousand suggests that varices are likely. The albumin-bilirubin score, called ALBI, has also been studied: a value below minus two point seven predicted a higher chance of live birth. In biliary atresia specifically, a Japanese study by Takahashi in 2021 suggested that cholinesterase, platelet count and MELD trends identified the women who did badly.

The practical point for biliary atresia is that the MELD score will often look reassuring, because synthetic function is preserved. The score that matters is portal hypertension. The spleen and the platelet count tell the truth. The young woman in our case, with platelets of ninety-two thousand and a spleen of seventeen centimetres, almost certainly has varices, whatever her MELD score says.

Part twelve. A pregnancy plan

A pregnancy in a woman with biliary atresia should be managed jointly by hepatology and high-risk obstetrics, ideally in a centre where endoscopy is available at any hour. The plan can be thought of in five stages.

Before conception, counsel the woman about risk, calculate her scores, and provide reliable contraception until she is ready. Estrogen-containing contraception should be avoided in decompensated cirrhosis, and intrauterine devices and progestin-only methods are usually preferred. The European association recommends a screening endoscopy within one year before conception for any woman with cirrhosis or known portal hypertension.

For varices, high-risk varices should be banded, and a non-selective beta-blocker should be started or continued during pregnancy. Because beta-blockers can affect fetal growth, growth scans are sensible. Both the American and European guidance agree on this approach.

For imaging, look for a splenic artery aneurysm, review the anatomy of the Roux loop, and agree in advance how cholangitis will be managed if it occurs.

For delivery, the European guidance is that delivery should be for obstetric reasons. A caesarean section is not required simply because of varices, and a shortened second stage of labour, with assisted delivery, reduces the time spent pushing.

After delivery, watch closely for six to twelve months, because jaundice, cholangitis and decline can appear late, as the Japanese series showed. If a woman presents already pregnant and has never had a screening endoscopy, the European guidance prefers to do it in the second trimester.

Finally, some women with biliary atresia will already have had a transplant. For them, the usual advice is to wait at least a year after transplantation, and to stop mycophenolate well before conception, because it causes birth defects. The 2023 European guideline advises stopping it twelve weeks before conception, while the drug label in the United States specifies at least six weeks. Tacrolimus is compatible with pregnancy, with monitoring of blood pressure, kidney function, glucose and drug levels.

Part thirteen. Cancer in the native liver

Cancer in the native liver of a patient with biliary atresia is rare, but it happens, and it happens at both ends of life. At King's College Hospital, Hadzic and colleagues found that three of three hundred and eighty-seven infants diagnosed between 1990 and 2008, or about zero point eight percent, developed histologically proven hepatocellular carcinoma, at a median age of only two years. Two more children referred from elsewhere had cancer found in the explanted liver, one at one year and one at almost eighteen years. Only two of the five had a raised alpha-fetoprotein. All five were treated by transplantation and were well at follow-up.

The 2025 systematic review by Madadi-Sanjani and colleagues gathered thirty-seven publications describing fifty-seven malignancies in patients with biliary atresia. There were forty-two hepatocellular carcinomas, eight cholangiocarcinomas, five hepatoblastomas, and two other cancers, one of the bowel and one of the thyroid. When drawn as bars, the bar for hepatocellular carcinoma was more than five times longer than the bar for cholangiocarcinoma. These are reported cases, not incidence, and publication bias is certain.

The ages are revealing. Hepatocellular carcinoma was diagnosed at a median age of four years, but the range was from seven months to thirty-four years, and a case in a thirty-eight-year-old woman

has also been reported. Alpha-fetoprotein was raised in about two thirds of those with data, so a normal value does not exclude cancer. Cholangiocarcinoma was diagnosed at a median age of twenty-three and a half years, with a range from eleven to sixty-three years. It is a disease of young adults with a long-standing Roux loop, chronic biliary injury, stones or residual cysts. CA nineteen-nine was raised in all three cases where it was measured.

Part fourteen. Two tumours, two prognoses

The two cancers behave very differently. Hepatocellular carcinoma in biliary atresia is usually found in a cirrhotic, often jaundiced liver, and frequently it is found by chance in the explanted liver at transplant. Of the cases in the review, twenty-eight were alive without recurrence after transplantation. Cholangiocarcinoma, by contrast, usually presents with symptoms: a new stricture, a mass at the porta, rising tumour markers, or cholangitis that will not settle. Of the five patients with outcome data, four died, two after transplant and two on palliative chemotherapy. A 2019 case report by Nio and colleagues described a sixty-three-year-old woman who had a cyst-type, correctable form of biliary atresia treated in infancy; she developed an unresectable cholangiocarcinoma at the porta and died before her sixty-fourth birthday.

This asymmetry is the strongest argument for surveillance. Transplantation cures most hepatocellular carcinomas found within the liver, whereas cholangiocarcinoma is usually found too late. The aim of surveillance is to catch the first early and to think of the second whenever the biliary picture changes.

Part fifteen. Surveillance for life

There is no guideline written specifically for cancer surveillance in biliary atresia, so practice borrows from adult cirrhosis guidance. The 2023 American association guidance on hepatocellular carcinoma recommends ultrasound with alpha-fetoprotein every six months in patients with cirrhosis. The European association guideline on hepatocellular carcinoma recommends ultrasound every six months and treats alpha-fetoprotein as optional. Since almost every adult with a native liver after a Kasai has cirrhosis, six-monthly ultrasound, with alpha-fetoprotein, is a reasonable standard for life.

The 2025 review by Madadi-Sanjani proposed going further: ultrasound and alpha-fetoprotein every three to six months from childhood, especially in children whose jaundice never fully cleared, and a yearly CA nineteen-nine from adolescence onwards to look for biliary cancer. This is a proposal from a review, not a guideline, but it is sensible. When ultrasound is poor because of adhesions, body habitus or a small, shrunken liver, contrast-enhanced magnetic resonance imaging is a better tool. A new stricture, a new mass or cholangitis that does not settle in an adult should prompt magnetic resonance imaging with cholangiography, tumour markers and, where possible, tissue diagnosis, together with early discussion with the transplant team. And every transplant candidate should have cross-sectional imaging at listing so that an unsuspected tumour is staged before the operation.

Part sixteen. What the guidelines say

When the guidelines are compared, the picture is one of partial agreement and important gaps. On structured transition, the guidance is aligned: the 2018 joint position paper of the European

paediatric society and the European association recommends it, and the 2013 guidance from the American association and the American Society of Transplantation on long-term care after paediatric liver transplantation supports it. On variceal screening around pregnancy, the American and European guidance are aligned: screen before or during pregnancy, with the European guideline specifying within one year before conception or otherwise in the second trimester. On variceal treatment in pregnancy, they are aligned again: beta-blockers or banding for high-risk varices.

There are differences in detail. On mycophenolate before conception, the European guideline advises stopping twelve weeks before, while the United States label requires at least six weeks. On surveillance for hepatocellular carcinoma, the American association recommends ultrasound with alpha-fetoprotein, while the European association recommends ultrasound and considers alpha-fetoprotein optional. The 2017 North American and European guideline on the cholestatic infant does not address adults at all.

There are also material gaps. No guideline addresses surveillance for cholangiocarcinoma after a Kasai, and no allocation system gives adults with biliary atresia a listing priority that matches their real risk. In these areas practice rests on case series and expert opinion.

Part seventeen. A stepwise plan and the common traps

A yearly review of an adult with a native liver after a Kasai can be built in five steps. First, check bilirubin, albumin, INR, platelet count and creatinine, and cholinesterase where it is available. Second, perform ultrasound and alpha-fetoprotein twice a year and CA nineteen-nine once a year, together with liver stiffness and spleen size. Third, manage varices according to the Baveno criteria, with endoscopy, beta-blockers or banding as needed, and image for splenic artery aneurysm. Fourth, check oxygen saturation, arrange an echocardiogram if there is breathlessness, and review vitamins, bone health, nutrition and mood. Fifth, discuss contraception or a pregnancy plan, review adherence, and state clearly the threshold for transplant referral. Every clinic letter should record the date and anatomy of the Kasai, the number of cholangitis episodes in the past year, the variceal status, the surveillance results, and whether a transplant centre knows the patient. If the answer to the last question is no, the patient should be referred.

There are common traps. The first is discharging a patient with a normal bilirubin, when cirrhosis and portal hypertension can progress silently. The second is ordering an ERCP for cholangitis after a Kasai, when the ducts drain into a Roux loop and the right path is imaging and interventional radiology. The third is being reassured by a MELD of nine, when cholangitis, varices and sarcopenia are not in the score. The fourth is transferring a young person during a crisis or by letter alone. The fifth is assuming that pregnancy is either forbidden or routine, when it is possible but needs a plan. The sixth is skipping endoscopy because the woman feels well. The seventh is missing a splenic artery aneurysm before pregnancy or surgery. The eighth is calling a new hilar stricture in a thirty-year-old benign without checking tumour markers and imaging for cholangiocarcinoma. The ninth is stopping surveillance after childhood, when hepatocellular carcinoma has been reported into the fourth decade. And the tenth is waiting for parents to donate, when adults often have no living donor and should be listed early.

Part eighteen. What to do differently in clinic

Let us return to the nineteen-year-old woman. Her bilirubin is above twenty-one micromoles per litre, she had cholangitis in adolescence, and her platelet count and spleen size tell us that she has portal hypertension. On the King's data, each of these features independently raises her chance of needing a transplant as an adult, so a transplant centre should know her now, before she is ill. Her transfer should happen in a joint clinic while she is well, with a written summary of her Kasai, her Roux loop, her cholangitis history and her varices. Before she tries to conceive, she should have an endoscopy, with banding of any high-risk varices and a beta-blocker, cross-sectional imaging to look for a splenic artery aneurysm, and a clear plan for cholangitis. Her pregnancy should be managed by a joint hepatology and high-risk obstetric team in a centre with endoscopy, with delivery for obstetric reasons and close follow-up for a year afterwards. And she should have ultrasound and alpha-fetoprotein every six months for the rest of her life with her native liver, with a yearly CA nineteen-nine, and a low threshold to investigate any new stricture.

The broader changes for clinic are these. Treat a cleared Kasai as a chronic disease, not a cure. Write the bilirubin at sixteen, the varices and the cholangitis history into every transfer letter. Never send an adult with biliary atresia for an ERCP without first thinking about the Roux loop. Do not let a low MELD score delay a transplant referral. Plan pregnancy before conception rather than reacting to it. And keep watching the native liver for cancer, for life, remembering that a new stricture in an adult is a tumour until proved otherwise. The adult with biliary atresia belongs to two specialties at once, and the most important thing we can do is to make sure that both of them know it.